Cosmic Ray Muons

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The main component of cosmic rays at sea-level and underground is represented by muons. Information about parameters describing strong and electromagnetic interactions can be extracted from measurements of muons and their interactions. This is of interest especially at high energies which are presently not available at accelerators. Furthermore muon fluxes and energy spectra of muons are important geophysical quantities.

1. Sources and Properties of Muons

Muons are normally generated in the decay of secondaries from proton interactions. Muons are unstable. They decay with a lifetime of $\tau = 2.2 \cdot 10^{-6}$ sec into electrons and neutrinos according to $\mu^+ \to e^+ + \nu_e + \bar{\nu}_\mu$, $\mu^- \to e^- + \bar{\nu}_e + \nu_\mu$
The muon mass is $m_\mu = 207\, m_e = 105.6594$ MeV and the muon is a lepton, i.e. its spin is $s = \frac{1}{2}\hbar$. As a Dirac particle the magnetic moment of the muon is given by

$$\mu_0 = \frac{e\hbar}{2 m_\mu c} .$$

The consideration of interactions with zero-point electromagnetic fields gives a more accurate value for the magnetic moment of the muon

$$\mu = \mu_0 \left(1 + \frac{\alpha}{\pi} + 0.75 \frac{\alpha^2}{\pi^2}\right) = 1.0011654 \mu_0 \quad ; \alpha = \frac{1}{137}$$

The deviation of the gyromagnetic ratio of the muon from 2 can be represented by $a_\mu = (g_\mu - 2)/2 = 0.0011654$ and $g_\mu = 2 \cdot 1.0011654$, where the gyromagnetic ratio is expressed in units of $e/2m_\mu c$.

The muon has to be characterized by an additive quantum number, a muonic lepton number, which distinguishes if from a mere heavy electron. The leptons have the following quantum numbers:

particle	$e^+, \bar{\nu}_e$	e^-, ν_e	$\mu^+, \bar{\nu}_\mu$	μ^-, ν_μ
lepton number	$+L_e$	$-L_e$	$+L_\mu$	$-L_\mu$

For this reason the reactions
$\mu^+ + e^- \to \mu^- + e^+$ and $\mu^\pm \to e^\pm + \gamma$ are forbidden.

2. Electron-Muon Universality

Muon and electron have many features in common. They are identical in the following properties.

a) spin
b) Fermi-Dirac statistics
c) strong interactions absent
d) electromagnetic and weak interactions present
e) cross-section for weak interactions
f) charge and sign of charge
g) no neutral particles
h) gyromagnetic ratio ($g_e = 2 \cdot 1.0011596$; $g_\mu = 2 \cdot 1.0011654$)
k) associated neutrinos, however $\nu_e \neq \nu_\mu$.

The only marked difference between these two leptons is their mass difference $m_\mu = 207\, m_e$. The fact that the associated neutrinos of muon and electron are different is perhaps a hint that the muon is not merely a heavy electron. From the theoretical point of view no arguments against the existance of other charged leptons are presented. A larger family of leptons is already taken into consideration. The decay modes of a heavy lepton μ^{*} could be $\mu^{*-} \to \mu^- + \bar{\nu}_\mu + \nu_{\mu^*}$ or $\mu^{*-} \to e^- + \bar{\nu}_e + \nu_{\mu^*}$ or $\mu^{*-} \to \pi^- + \nu_{\mu^*}$.
A serious difficulty in the experimental discovery of such heavy leptons is presented by its rapidly decreasing lifetime with increasing mass according to

$\tau_{\mu^*} \sim m_{\mu^*}^{-5}$ (e.g. for $m_{\mu^*} \approx$ 1 GeV follows $\tau_{\mu^*} \approx 10^{-11}$ sec). So far no heavy leptons* have been discovered in accelerator and cosmic ray experiments.

*inzwischen überholt

3. Muon Physics in Cosmic Rays

Muon physics at accelerators has been reviewed by Weissenberg 1967 and Toner 1969 and this is the reason why we deal here only with muons in cosmic rays.

Muons constitute the main component of cosmic rays at sea-level and are apart from neutrinos and secondaries the only component underground. Therefore the absolute intensity of muons, their energy spectrum and charge ratio represent important geophysical quantities. The measurement of these quantities yields indirect information on strong interactions at energies beyond the energy region presently available at accelerators (> 2 TeV $= 2 \cdot 10^{12}$ eV), since muons are tertiary products from proton interactions in the upper atmosphere. Using the known values for multiplicity and inelasticity in strong interactions a sea-level muon of 5 TeV corresponds typically to a primary proton of roughly 100 TeV.

3.1 Electromagnetic Interactions of Muons

The different types of electromagnetic interactions of muons which are investigated in cosmic rays are the scattering on electrons (knock-on process), the direct pair production of electrons, the bremsstrahlung and nuclear interactions. The latter represents multiple hadron production ($\rho, n, \pi, K, \ldots$) via the virtual photon spectrum of the muon.

There is interest in the knowledge of the dependence of these cross-sections on the muon energy and the energy transferred to the secondaries in the interaction $\sigma(E_\mu, E')$. The measurement of these cross-sections can be done with a combined setup of a muon spectrometer, in which the momentum of the muon is determined, and a calorimeter in which the muons undergo interactions and the energy of the secondaries is measured. For the first three types of interactions the secondaries (electrons or photons) will initiate cascades. However, the initial signature of the interactions will be obliterated by this cascade so that an experimental discrimination of these three interactions is not practicable. But this can be done from a theoretical point of view due to the different dependence of the cross-sections on muon energy and energy transfer. An example for the dependence of these interactions on the energy transferred to the secondaries at an average muon energy of E_μ = 1000 GeV along with experimental results is presented in fig.1 (Grupen 1972).

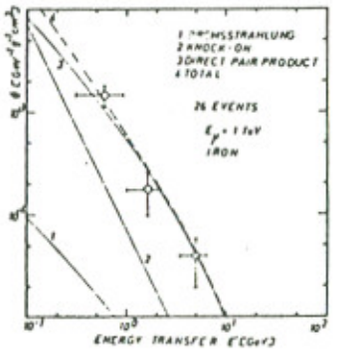

Fig.1: Electromagnetic Interactions of 1 TeV Muons

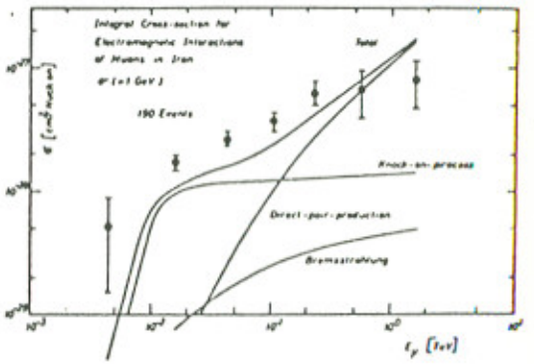

Fig.2: Integral Cross-Section for Electromagnetic Interactions of Muons

The dependence of the integral cross-section $\sigma(>E')$ for energy transfers in excess of 1 GeV as a function of the muon energy is plotted in figure 2 (Allkofer and Grupen 1972a). The experimental errors do not allow to draw firm conclusions about the validity of current theories on electromagnetic interactions

at high muon energies and large energy transfers.

The nuclear interaction of the muon can be separated experimentally from the others by pattern recognition as far as multiple charged hadron production is concerned. This is based on the different development of electromagnetic and hadronic cascades. Neutron production by muons, the other type of nuclear interaction, however, requires a different experimental set up to be detected.
For the measurement of neutron production by muons the calorimeter has to be replaced by a neutron monitor which sits behind the muon spectrograph. The investigation of nuclear interactions of the muon gives indirect information on the photonuclear cross-section for hadron production. The result of $\sigma_{ph} \approx 10^{-28} cm^2$/nucl. up to 100 GeV virtual photon energy is consistent with accelerator data.

A direct consequence of the electromagnetic interactions of muons is the energy-loss-relation which can be approximated to $-\frac{dE}{dx} = a + bE$ where a describes the energy loss due to muon scattering on electrons ($a \approx 2$ MeV/g/cm^2) and b is composed of the loss due to pair production ($b_{pp} \approx 1.8 \cdot 10^{-6} g^{-1} cm^2$), bremsstrahlung ($b_b \approx 2.1 \cdot 10^{-6} g^{-1} cm^2$) and nuclear interactions ($b_{n\nu} \approx 0.3 \cdot 10^{-6} g^{-1} cm^2$).

The knowledge of the cross-sections for electromagnetic interactions enables a determination of the muon spectra at very high energies (> 5 TeV) by measuring the depth-intensity relation of muons underground. The depth-intensity relation can be converted into a sea-level muon spectrum by an appropriate consideration of the energy-loss of muons underground. On the other hand a known sea-level muon spectrum would give information on the electromagnetic cross-sections (via the energy loss parameters a and b) of muons by comparison with the depth-intensity curve.

The results of the measurement of electromagnetic interactions can be compared with related theories and comparison can be made with equivalent accelerator experiments. Significant deviations could reveal imperfect theories or differences between cosmic ray and accelerator experiments could be explained in terms of new particle or new interactions.

A summary of the present state of electromagnetic interactions in cosmic rays was given by Allkofer et al. 1968 and Allkofer et al. 1971a.

3.2 Sea-level Muon Spectra and Charge Ratio

During the propagation of primary cosmic rays through the atmosphere secondary particles are produced and the main component at sea-level is constituted by muons (80 % of the total flux).
The spectrum of primary protons and heavy nuclei of the form

$$J_p(E) = const \cdot E^{-\gamma_p} \quad \text{where} \quad 2 \leq \gamma_p \leq 3 \qquad \gamma_p \simeq 2.4$$

produces a pion spectrum of the shape $J_\pi(E) = const \cdot E^{-\gamma_\pi}$.

It is a reasonable assumption that all pions are generated at an atmospheric depth of 100 g/cm^2 and decay on a very short distance into muons. The mean path of pions can easily be calculated to

$$s = v \cdot \tau_\pi \approx c \cdot \gamma \cdot \tau_\pi^0 = \frac{p}{m_\pi} \cdot \tau_\pi^0$$

Using a production altitude of 16 km where the air density is $\rho = 1.65 \cdot 10^{-4}$ g/cm^3 we arrive at s(2 GeV) = 110 m or 1.82 g/cm^2, s(200 GeV) = 11 km or 182 g/cm^2

Since the interaction length of pions is about $\lambda = 80$ g/cm^2 the 2 GeV pion will decay on a very short distance and the 200 GeV pion will interact.

The sea-level muon spectrum can be obtained from the pion source spectrum by folding with the survival probability of muons

$$N_\mu(E) = const \; P_\mu(E, t_0) \; (E + \Delta E_0)^{-\gamma_\pi}$$

where $\gamma_\pi \approx 2.8$ and $\Delta E_o \approx 2$ GeV (Energy loss of the muons in the atmosphere) and $P_\mu(E,t_o)$ is the probability of a muon produced at $t_o = 100$ g/cm^2 surviving to sea-level.

Some typical values for survival probabilities of muons produced at $t_c = 100$ g/cm^2 are given below.

E [GeV]	0.4	10	100
P [%]	20	80	97

There is a variety of ways to measure muon spectra. The most accurate and most direct way of doing this is given by spectrographs in which the deflection of the muon in a magnetic field is determined by aid of track detectors (see fig.3). The momentum of the muon is calculated by $p = \frac{300}{\Theta}\int B d\ell$, where $\int B d\ell$ is the magnetic field line integral.

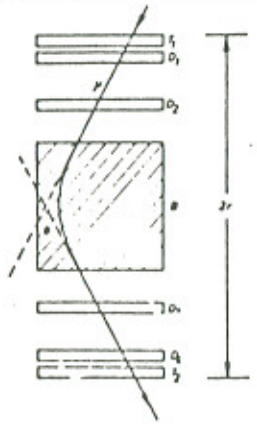

Fig.3: Sketch of a Magnetic Spectrograph

D_i (i=1,2,3,4) are track detectors

B is the magnetic induction in the solid iron stack

Θ is the deflection of the incident muon

$2r$ is the overall height of the apparatus

T_1, T_2 - Trigger counters of area F

Such a magnetic spectrograph is characterized by two important features:

a) Maximum detectable momentum (mdm). This is the momentum where the error in momentum determination becomes 100 %, i.e. $\Delta p/p = \Delta\Theta/\Theta = 1$. The mdm is limited by the space resolution of the track detectors, the alignment of the spectrograph and multiple scattering of muons. Presently mdm values of about 5 TeV can be obtained

b) Collection power or acceptance ΩA (solid angle times area), which is a figure determining the number of muons which can be recorded per unit time. The acceptance is calculated from the geometry of the triggering system: $\Omega A = F^2/4r^2$. The operation of a spectrograph with an mdm of 5 TeV requires an acceptance of $\Omega A > 1000$ cm^{2}sr to record a sufficient number of particles in any reasonable time.

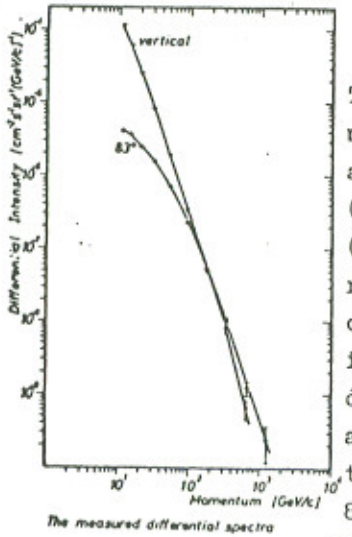

Fig.4: Muon Spectra at two Different Zenith Angles

Two typical muon spectra as measured in the vertical (0°) and near horizontal direction (83°) are plotted in fig.4 (Allkofer et al. 1971b). The reduced intensity of the inclined spectrum at low energies is due to the absorption and decay of low energy muons in the atmosphere. (The atmospheric thickness at 83° is about 8000 g/cm^2, i.e. about a factor of 8 larger than in the vertical direction.) The enhancement in intensity at high energies is caused by the favoured decay of the parent pions in rarer parts of the atmosphere against their interaction. This competition between decay and interaction leads to a secΘ enhancement at high energies (Θ -zenith angle).

Since positive and negative muons are deflected in different ways a magnetic spectrograph determines automatically the charge ratio of muons.

The positive character of primary cosmic rays is diluted by subsequent interactions in which particles of both signs are generated. It is obvious however

that a charge ratio of muons exceeding unity is to be expected. Assuming the equivalence of the two interactions

$$p + N \rightarrow p + N + n\pi^+ + n\pi^-$$

$$p + N \rightarrow n + N + (n+1)\pi^+ + n\pi^-$$

we have

$$\frac{N(\pi^+)}{N(\pi^-)} = \frac{2n+1}{2n} = 1 + \frac{1}{2n} .$$

Assuming further that all pions decay into muons the pion charge ratio equals the muon charge ratio. This simple model would lead to a charge ratio of $N(\mu^+)/N(\mu^-)$ = 1.25 for low muon energies where the corresponding multiplicity in the proton interaction is $2n = 4$. This prediction is already in close agreement with observation.

However, a dilution of the muon charge ratio is expected for the following reasons:

a) Primary cosmic rays are not only composited of protons but also by heavier nuclei which contain neutrons.
b) Secondary nucleons (p, n) are produced also in primary collisions. These will not lead to a positive charge excess.
c) At low momenta where the charge ratio is high due to the low multiplicity, muons are lost by decay before reaching sea-level.
d) High energy pions will interact and increase the effective multiplicity.

The momentum dependence of the charge ratio reflects the various interaction processes of the parent component. Depending on the parameters describing these processes different theoretical functions for the charge ratio are obtained. The statistical accuracy of presently available data does not allow to discriminate between various theoretical predictions. The experiments are not inconsistent with a constant value of $N(\mu^+)/N(\mu^-)$ = 1.28 over the entire energy

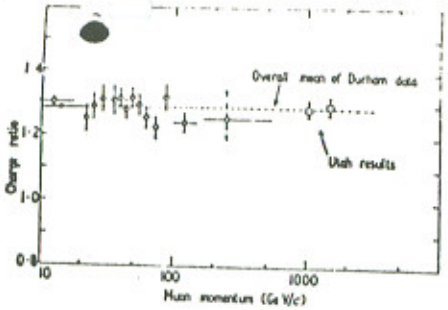

Fig.5: Charge Ratio of Muons

range up to some TeV (see figure 5; Ayre et al. 1972). A constant charge ratio is predicted by scaling ideas, however, the absolute value of the prediction is in disagreement with observation (Frazer et al. 1972).

3.3 Burst Measurements

An independent method of the determination of muon spectra is given by the burst - or burst-size method. A burst is an electromagnetic or hadronic cascade induced by a muon. A burst calorimeter is a stacked assembly of targets and particle detectors. The burst calorimeter (see Fig.6) is triggered by muons. These muons will undergo occasionally interactions in the absorbers. The energy of the secondaries is determined from the development of the shower in the burst array.

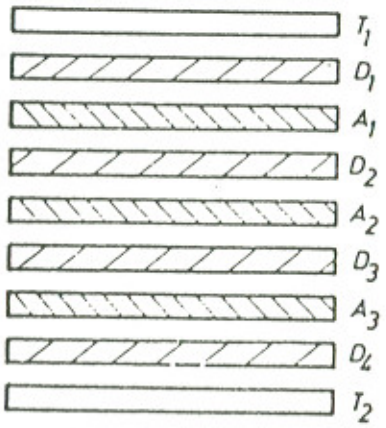

Fig.6: Burst-Array

(T_i - Trigger counters
D_i - Detectors
A_i - Absorbers)

Let $P(E_\mu, E')$ be the total probability for an energy transfer E' by a muon of energy E_μ, then the quantity which is measured in the burst array is

$$P(E')dE' = \int_{E_\mu} N(E_\mu) P(E_\mu, E') dE_\mu .$$

$P(E_\mu, E')$ is related to the cross-section of electromagnetic interactions.

Assuming the muon spectrum to be of the fo
$N(E_\mu)dE_\mu = k \cdot E_\mu^{-\gamma} dE_\mu$ this spectrum can be obtained from the measured quantity P (E') by a fitting procedure.

The burst-size technique uses only the reduced information of basically one detector covered with target in which the number of penetrating particles is recorded. It is believed that this number of particles S^*, the burst size, is part of a shower initiated by a muon. The measured quantity in this case is

$$\phi(\geq S) = \int_{S^*} \int_{E'} \int_{E_\mu} N(E_\mu)\, P(E_\mu, E')\, R(E', S^*)\, dE_\mu\, dE'\, dS^*$$

where $R(E',S^*)$ is the probability that a secondary particle of Energy E' produces a burst of size S^* at the depth of observation. The measured quantity, the integral burst-size spectrum, is related to the muon spectrum and this again can be obtained in a similar way as from the burst spectrum.

It is obvious that the accuracy of this method of the determination of muon spectra suffers from various disadvantages; in particular the cross-sections for electromagnetic interactions are assumed to be well-known. A determination of the absolute intensity of muons at high energies cannot be expected from this technique due to large fluctuations in burst-sizes. However, the slope of the muon spectrum up to some 10 TeV could be determined using this method.

3.4 Absolute Intensity of Sea-Level Muons

The absolute intensity of sea-level muons represents a kind of natural constant of cosmic rays. The knowledge on sea-level muons spectra as obtained by range measurements and magnetic deflection of muons were combined by Rossi 1948 to form a standard integral spectrum. To get an absolute spectrum this standard inte[illegible] muon spectrum was normalized to the absolute integral intensity as measured previously by Greisen at the momentum of 0.2 GeV/c. On the basis of this spectrum the differential spectrum was calculated and the absolute differential intensity at 1 GeV/c, the so-called 'Rossi-value', fixed. Later measurements on muon spectra used this Rossi-value as a normalization point. It is not surprising that a remeasurement of the differential intensity at 1 GeV/c gave a somewhat different value which is about 25 % higher than Rossi's value. (The absolute differential intensity at 1 GeV/c is $I=(3.09\pm0.25)\cdot 10^{-}$ $[cm^{-2}sec^{-1}sr^{-1}\ (GeV/c)^{-1}]$; Allkofer and Jokisch 1972b) This determinatio of the absolute intensity can easily be done with a range spectrometer (see fig.7) by measuring the signal $D_1\ D_2\ D_3\ D_4$ which is the number of muons having penetrated the absorber A_1 (corresponding to an energy loss E) and stopped in A_2 ($\hat{=}\ \Delta E$). This counting rate gives the differential intensity at the energy E. The threefold coincidence $D_1D_2D_3$ gives the absolute integral intensity greater E For a survey on absolute intensities see Allkofer and Jokisch 1972 b.

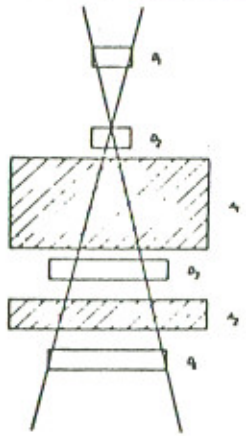

Fig.7: Range Spectrometer

3.5 Muons Underground and Zenith Angle Dependence

It has already been mentioned that the measurement of muons underground enables a determination of the muon spectrum in the TeV range via electromagnetic interactions of muons. The energy loss relation $-\frac{dE}{dx} = a + bE$ holds approximately for muons and gives a mean range by integration

$$R = \int dx = \int \frac{dE}{a+bE} = \frac{1}{b} \log\left(\frac{a+bE}{a}\right)$$

or

$$E = \frac{a}{b}\,(e^{bx} - 1)$$

For low energy muons ($E \ll 100$ GeV) with $b \ll a$ these relations reduce to $R = E/a$ and $E = aR$ respectively. Current numerical values for a and b are $a \approx 2$ MeV/g/cm^2; $b \approx 4 \cdot 10^{-6}$ g^{-1}cm^2. The zenith angle dependence of muons can be approximated by $j(x,\Theta) = j(x,0) \cos^m \Theta$.
For sea-level (x = 0) we have m = 2. The value of m increases rapidly with increasing depth underground and for very large depths a $\cos^m \Theta$ dependence is insufficient to describe the angular distributions of muons which are highly peaked about the vertical direction.

If the integral muon spectrum is described by $E_\mu^{-\gamma}$ the depth-intensity relation is

$$j(x,\Theta) \sim \sec\Theta \exp\{-\gamma b x \sec\Theta\}.$$

Experimental observations on the intensity of muons underground have been extended up to 10 000 m.w.e.

3.6 Strong Interactions

The shape of the muon spectrum, its absolute intensity and the charge ratio reflect the properties of strong interactions of primary cosmic rays. The primary spectrum propagates through the atmosphere producing a spectrum of secondary mesons (pion and kaon spectrum). These secondary spectra serve as source for muons which are observed at sea-level. The sea-level muon spectrum can thus be predicted from the measured primary cosmic ray spectrum. Into this prediction enter assumptions on properties of hadronic interactions such as inelasticity, multiplicity, mean interaction length, pion to kaon ratio, production of hyperons in energetic collisions. These interactions are governed by conservation laws of energy, momentum, strangeness, parity etc. The comparison between predicted muon spectra, absolute intensities and charge ratios at sea-level and the corresponding observed quantities provides a means to check whether the interaction characteristics as obtained at accelerator energies still hold at cosmic ray energies. Deviations could be interpreted in terms of modifications of known parameters or the postulation of new types of interactions or particles.

In an equivalent way the primary cosmic ray spectrum can be calculated from the measured sea-level muon spectrum. If the muon spectrum is known up to about 5 TeV, the primary proton spectrum can be deduced up to energies as high as 100 TeV.

The best agreement between known muon spectra and primary spectra is obtained with the following set of parameters (Jabs 1968):

collision mean free path of nucleons	$\lambda_N = 70$ g/cm^2
collision mean free path of pions	$\lambda_\pi = 74$ g/cm^2
mean total inelasticity	$K = 0.40$
kaon to pion ratio	$K/\pi = 0.20$
multiplicity dependence	$n \sim \log E$

Further experimental results about muons in cosmic rays and discussions of new phenomena in cosmic rays have been reviewed by Allkofer 1972c,d.

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